UNIT 4 · TOPIC 1(B)

Fuels and Combustion in Farm Engines

Syllabus topic: Engineering thermodynamics, power cycles and fuels

Diesel (HSD) — the Indian farm fuel

Property	Value / meaning
Calorific value (CV)	≈ 42–45 MJ/kg
Density	820–860 kg/m³
CETANE number	Ignition quality (short ignition delay) — BS-VI HSD minimum **51**
Sulphur	≤ **10 ppm** (BS-VI nationwide since April 2020)
Flash point	> 35 °C (safety)
Cloud / pour points	Matter in cold weather
Viscosity	Lubricates the injection pump

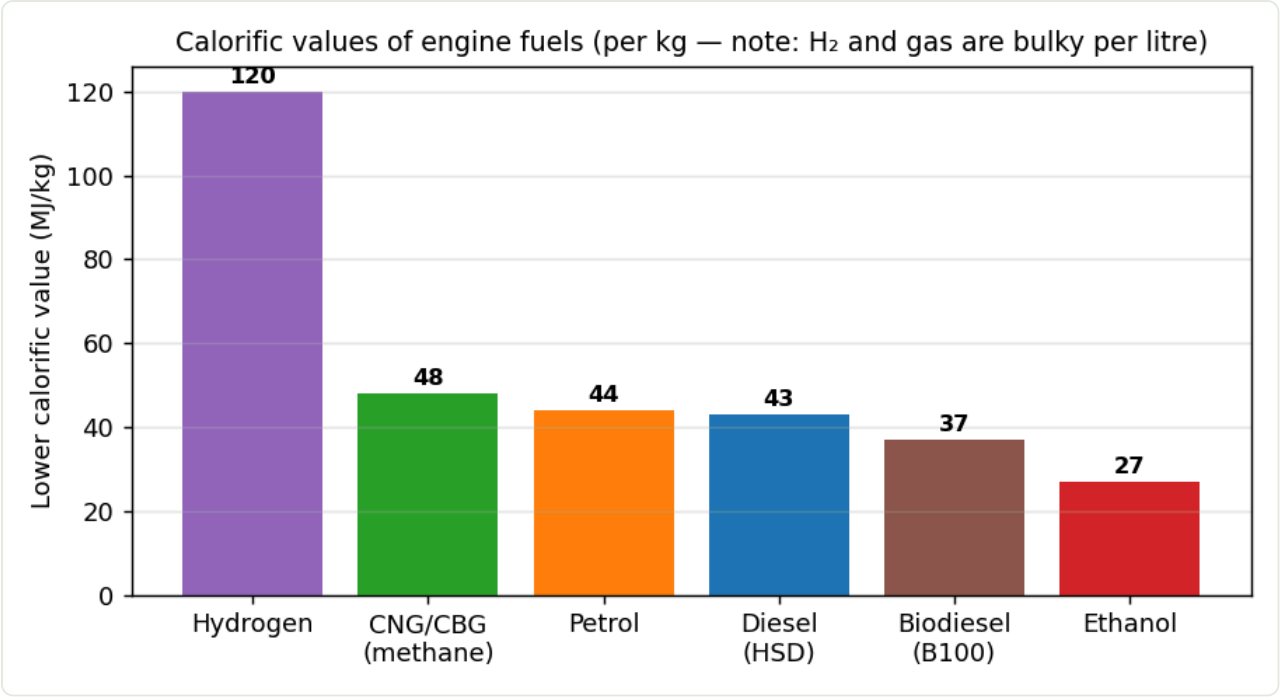
- Stoichiometric A/F ≈ **14.5:1** by mass, but diesels run LEAN (**18–70:1** across the load range).
- Diesels are unthrottled — load is governed by the fuel quantity.

Petrol and the octane–cetane see-saw

- **OCTANE number** = resistance to knock (iso-octane = 100, n-heptane = 0). Regular Indian petrol: 91 RON.
- High octane ↔ LOW cetane, and vice versa — a good SI fuel is a bad CI fuel.
- **Fuel-quality tests:** distillation range, RVP (volatility), gum content, calorific value (bomb calorimeter — Unit 9).

Update: E20 petrol sold from 1 April 2026 must meet a minimum RON of 95 (MoPNG notification, 17 Feb 2026).

Alternative and renewable fuels (details in Unit 7)



Calorific values of engine fuels per kg (hydrogen and gases are bulky per litre)

Fuel	Key facts
Ethanol	E20 / E85 / ED95
Biodiesel	B5–B20; transesterified methyl esters; CV ~37 MJ/kg; cetane 50+
CNG / CBG	Methane, ~48 MJ/kg; SATAT programme
LPG, producer gas	Gaseous alternatives
Hydrogen	120 MJ/kg — H_2-ICE and fuel cells
Battery-electric	E-tractors (Topic 7b)

Blends need material compatibility (elastomers), and the engine must be recalibrated to offset the CV drop.

Combustion in engines

- **SI:** a premixed flame from the spark. Abnormal combustion: knock (end-gas auto-ignition) and surface ignition.

CI combustion sequence:

1. Ignition DELAY period.
2. Premixed rapid burn — diesel knock if the delay is long (LOW cetane worsens it).
3. Diffusion burn.
4. Afterburn.

Cold-starting aids: glow plugs, higher cranking speed, ether.

Updated information — fuels (verified July 2026 in the source notes)

Item	Status
Ethanol blending	India reached 20% blending (E20) in November 2025 — five years ahead of the original 2030 target. E20 became MANDATORY across all States/UTs from **1 April 2026**.
Ethanol supply	38 crore L (2014) → ~661 crore L (June 2025)
Roadmap	E22 next; E30 in the 2028–2030 window (PIB/MoPNG)
Fuel quality	BS-VI (10 ppm sulphur) auto fuels nationwide since April 2020; HSD cetane minimum 51

ARS/NET increasingly asks current-affairs-linked engineering (E20 date, TREM stages, PM limits). These facts are current as of July 2026 — recheck close to the exam. Emission norms: Topic 7a.

Practice MCQs

Q19. Cetane number measures:

(a) Knock resistance of petrol
(b) Ignition quality (short delay) of diesel
(c) Volatility
(d) Viscosity

Q20. Minimum cetane number of BS-VI Indian HSD is:

(a) 40
(b) 45
(c) 51
(d) 60

Q21. BS-VI fuel sulphur limit is:

(a) 10 ppm
(b) 50 ppm
(c) 350 ppm
(d) 500 ppm

Q22. Calorific value of diesel is about:

(a) 27
(b) 35
(c) 42–45
(d) 55 MJ/kg

Q23. A fuel with HIGH octane number has cetane number that is:

(a) High
(b) Low
(c) Same
(d) Unrelated

Q24. Octane rating scale uses iso-octane = 100 and:

(a) Benzene = 0
(b) n-heptane = 0
(c) Cetane = 0
(d) Methane = 0

Q25. Diesel engines control load by:

(a) Throttling air
(b) Varying fuel injected into unthrottled air (lean overall)
(c) Spark timing
(d) Valve lift only

Q26. 'Diesel knock' results from:

(a) Too short ignition delay
(b) Too long ignition delay (low cetane)
(c) High octane
(d) Rich mixture

Q27. Calorific value of fuels is measured with a:

(a) Patternator
(b) Bomb calorimeter
(c) Dynamometer
(d) Hydrometer

Q28. Hydrogen's calorific value per kg is roughly:

(a) Equal to diesel
(b) Half of diesel
(c) Nearly 3× diesel (~120 MJ/kg)
(d) 10× diesel

Q29. Biodiesel is produced from vegetable oils by:

(a) Distillation
(b) Transesterification
(c) Cracking only
(d) Fermentation

Q30. Ethanol blending of 20% (E20) became mandatory across India from:

(a) April 2020
(b) April 2023
(c) April 2026
(d) 2030

Q31. India achieved the 20% average ethanol-blending milestone in:

(a) 2020
(b) 2023
(c) November 2025
(d) Not yet

Answers

Q19 — (b) Ignition quality (short delay) of diesel

Q20 — (c) 51

Q21 — (a) 10 ppm

Q22 — (c) 42–45

Q23 — (b) Low

Q24 — (b) n-heptane = 0

Q25 — (b) Varying fuel injected into unthrottled air (lean overall)

Q26 — (b) Too long ignition delay (low cetane) — *A long delay lets too much fuel accumulate → sudden premixed burn → pressure shock; cured by higher cetane, a warm engine, higher r.*

Q27 — (b) Bomb calorimeter

Q28 — (c) Nearly 3× diesel (~120 MJ/kg)

Q29 — (b) Transesterification

Q30 — (c) April 2026 — *From the updated information box (PIB/MoPNG; MoRTH TREM notifications; verified July 2026).*

Q31 — (c) November 2025

Sources: Liljedahl, Turnquist, Smith & Hoki — Tractors and their Power Units; standard thermodynamics texts; PIB/MoPNG releases on ethanol blending; MoRTH/CMVR TREM notifications and DieselNet (as verified in the source notes, July 2026); NPTEL lectures. MoPNG notification of 17 Feb 2026 (E20, RON 95).